Yes. For tomorrow, you **do not need to know every technical detail**. You need to understand the project story and be able to explain the important concepts simply.

1. Your project in one sentence

> **We are using machine learning to detect unusual behaviour in LoRaWAN traffic and identify which node may be behaving like a rogue node**.

---

2. What is LoRaWAN?

LoRa

A wireless technology that allows devices to communicate over **long distances using low power**.

LoRaWAN

The **network protocol** that uses LoRa for IoT communication.

Simple difference:

> **LoRa = communication technology**
> **LoRaWAN = network system/protocol using it**

---

3. Basic LoRaWAN architecture

Remember:

```
IoT Device
    ↓
 Gateway
    ↓
Network Server
    ↓
Application Server
```

IoT Device

Sensor/node that sends data.

Gateway

Receives wireless LoRa messages and forwards them.

Network Server

Manages the LoRaWAN network and devices.

Application Server

Handles the actual application data.

---

4. What is a rogue node?

A rogue node is a node that behaves abnormally or suspiciously.

For example:

```
Normal node → sends data normally
```

```
Rogue node → sends unusual/frequent/abnormal traffic
```

Your goal is:

```
Traffic
   ↓
Find abnormal behaviour
   ↓
Find which device is responsible
   ↓
Potential rogue node
```

---

5. What is anomaly detection?

An anomaly is something that doesn't look normal.

Example:

If a sensor normally sends:

```
1 packet every 30 seconds
```

and suddenly sends:

```
100 packets every second
```

that behaviour is unusual.

Your project tries to automatically detect such behaviour.

---

6. What are your features?

Your model looks at characteristics of the traffic.

RSSI

How strong the received signal is.

SNR

How good/clean the signal is compared to noise.

SF — Spreading Factor

A LoRa communication setting.

Higher SF generally means:

- longer range
- slower communication
- more airtime

Inter-arrival time

Time between two packets.

Example:

```
Packet 1 → 10:00:00
Packet 2 → 10:00:20

IAT = 20 seconds
```

FCnt — Frame Counter

A counter associated with LoRaWAN frames.

It can help identify unusual counter behaviour, but raw FCnt can also create problems in our current synthetic dataset, which we'll discuss below.

---

7. Why use multiple features?

Because one feature isn't enough.

For example:

> Low RSSI does NOT automatically mean rogue.

A legitimate device could have low RSSI because it is far away.

So we combine:

```
RSSI
+
SNR
+
SF
+
Packet timing
+
FCnt behaviour
```

```
        ↓
ML model
```

---

8. What is an IDS?

IDS = Intrusion Detection System

It detects suspicious activity in a network.

Two simple types:

Signature-based

Looks for known attack patterns.

```
Known attack pattern?
       ↓
     Yes → Alert
```

Anomaly-based

Learns what normal behaviour looks like.

```
Normal behaviour
       ↓
New traffic
       ↓
Looks very different?
       ↓
Anomaly
```

Your project mainly focuses on anomaly detection.

---

9. Why anomaly detection?

Because you may not know every possible attack beforehand.

Instead of asking:

> "Is this exactly Attack X?"

we ask:

> "Does this behaviour look abnormal?"

This is particularly useful for unknown or previously unseen behaviour.

---

10. Your three models

You have:

```
Random Forest
XGBoost
Isolation Forest
```

You don't need to go extremely deep.

Random Forest

It uses many decision trees.

Think:

```
Tree 1 → Normal
Tree 2 → Rogue
Tree 3 → Rogue
Tree 4 → Rogue

Majority → Rogue
```

It is a supervised model, meaning it learns from labelled examples.

XGBoost

Also uses decision trees, but builds them one after another, with later trees trying to correct earlier mistakes.

Simple difference:

> Random Forest = many trees vote
>
> XGBoost = trees learn from previous mistakes

It is also supervised.

---

11. Isolation Forest

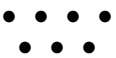
This one is important.

It tries to find points that are easy to separate from the rest.

Imagine:

```
Normal:
```

• • •



```
Anomaly:

              X
```

The X is easy to isolate.

So:

> Easy to isolate → likely anomaly

For a proper anomaly-detection experiment, we want:

```
Normal traffic
      ↓
Train Isolation Forest
      ↓
Test on:
 ┌─────────────────┐
 Normal + Rogue
 └─────────────────┘
```

The current repository does not yet follow this ideal setup, so describe it as a planned improvement.

---

12. Why three models?

Very simple answer:

> "Random Forest and XGBoost show how well supervised classification works when we have labels. Isolation Forest gives us an anomaly-detection approach where we can learn normal behaviour without needing every attack to be labelled."

---

13. Your current dataset

The current repository uses:

- **100,000 records**
- **5 benign nodes**
- **1 rogue node**
- **synthetic data**

The 5+1 setup is completely fine for your project.

You don't need to say 6 normal nodes.

---

14. What is the problem with the synthetic data?

This is important.

Currently, the generator makes normal and rogue traffic **too different**.

For example, roughly:

```
Normal:
RSSI = around -85

Rogue:
RSSI = around -115
```

and similarly for other features.

So the model can almost say:

> "Very low RSSI = rogue."

That's an easy classification problem.

---

15. Is synthetic data itself bad?

No.

Synthetic data is useful for creating controlled experiments.

The problem is when the synthetic data is too unrealistic or too easily separated.

Better approach:

```
Real / semi-real traffic
        ↓
Learn realistic patterns
        ↓
Generate synthetic data
        ↓
Add realistic anomalies
```

So yes, your idea of giving us semi-synthetic real-world data and generating better synthetic data from it is absolutely possible.

---

16. What is FCnt leakage?

This is another important issue.

Currently the synthetic generator gives the rogue node unusual FCnt values.

So the model can learn:

```
Weird FCnt → Rogue
```

instead of actually learning suspicious behaviour.

That's called a dataset shortcut / feature leakage problem.

Solution

We can:

- remove raw FCnt, or
- use derived features such as FCnt changes/resets/repetitions.

---

17. What is the Isolation Forest improvement?

Current idea:

```
All data
 ↓
Train Isolation Forest
```

Better:

```
NORMAL DATA ONLY
       ↓
Isolation Forest
       ↓
Test on mixed data
       ↓
Normal / Anomaly
```

Why?

Because we're trying to teach Isolation Forest:

> "This is what normal behaviour looks like."

Then we ask:

> "Does this new traffic look normal?"

---

18. Packet-level vs node-level

This is VERY important.

The current prototype mainly predicts:

```
Packet → Normal / Rogue
```

But your actual goal is:

```
Which NODE is rogue?
```

So eventually:

```
Packet predictions
        ↓
Group by device
        ↓
Count anomalous packets
        ↓
Anomaly percentage per node
        ↓
Suspicious node
```

Example:

```
Node A → 2% anomalous
Node B → 3%
Node C → 65%   ← suspicious
Node D → 1%
```

---

19. What does Streamlit do?

Streamlit is not the AI model.

It is the dashboard/interface.

```
ML model
   ↓
Predictions
   ↓
Streamlit
   ↓
Graphs + tables + results
```

Your current dashboard lets you select models and visualize predictions/traffic.

---

20. What is the biggest limitation of your current project?

If they ask:

> "What is the main limitation?"

Say:

> "The current synthetic data has distributions that are too clearly separated, so the classification problem is easier than a real-world problem. We also need to address raw FCnt leakage and improve the Isolation Forest training setup."

That's a very good answer.

---

21. What will you improve?

Remember these four:

1. Better data

Use real/semi-real LoRaWAN traffic to create more realistic synthetic data.

2. Reduce feature leakage

Especially raw FCnt.

3. Better Isolation Forest

Train it using normal traffic only.

4. Node-level detection

Aggregate packet anomalies by device to identify suspicious nodes.

---

22. Metrics you should know

Accuracy

How many predictions were correct overall.

Precision

> Of everything we called rogue, how much was actually rogue?

Recall

> Of all actual rogue traffic, how much did we detect?

F1

Balance between precision and recall.

False Positive

```
Normal → incorrectly called Rogue
```

False Negative

```
Rogue → incorrectly called Normal
```

For your project, precision, recall and F1 are more informative than accuracy alone.

---

23. Very likely questions + easy answers

"Why machine learning?"

Because traffic behaviour can be complex, and ML can learn relationships between multiple features instead of relying only on fixed thresholds.

"Why not just use RSSI?"

RSSI naturally changes due to distance, obstacles and interference, so it cannot identify a rogue node by itself.

"Why Random Forest?"

It works well with tabular data and provides a strong, simple baseline.

"Why XGBoost?"

It is another strong tree-based model and gives us a comparison with Random Forest.

"Why Isolation Forest?"

It is designed for anomaly detection and can learn normal behaviour without needing labels for every attack.

"Why synthetic data?"

It gives us controlled data and known labels for the initial prototype.

"Is your synthetic data realistic?"

Not completely. That is one of our current limitations, and we plan to improve it using real/semi-real traffic.

"Can your model detect every attack?"

No. Detection depends on whether the attack produces observable differences in the features we use.

"Can a legitimate node look anomalous?"

Yes. Wireless conditions naturally change, which is why multiple features and node-level/temporal analysis are important.

"Can you have two rogue nodes?"

Yes. The system can calculate anomaly behaviour for each node rather than assuming there is only one rogue node.

"What is your output?"

> Currently packet-level normal/anomaly predictions. The intended final output is suspicious-node identification through aggregation.

---

24. If they challenge your project

Faculty:

> "Your model gets very high accuracy. Why should I believe it?"

Answer:

> "High performance on the current synthetic data doesn't prove real-world performance. The current data is controlled and relatively easy to separate. That's why realistic data and proper validation are important next steps."

Faculty:

> "So your current model isn't reliable?"

Answer:

> "It demonstrates that the complete detection pipeline works, but we would not claim real-world reliability from the current synthetic experiment alone."

Faculty:

> "Then why did you build it?"

Answer:

> "The prototype lets us validate the complete pipeline first. Now we can improve the dataset and evaluation methodology rather than debugging the entire system at the same time."

Those are very strong answers.

---

25. Your project story — memorize this

If you remember nothing else, remember:

```
LoRaWAN network
      ↓
Nodes generate traffic
      ↓
Collect telemetry
      ↓
Extract features
      ↓
```

```
Machine Learning
      ↓
Detect abnormal packets
      ↓
Group by node
      ↓
Identify suspicious node
```

And:

> Current prototype = controlled synthetic data + RF/XGBoost/Isolation Forest + dashboard.
>
> Main improvements = realistic data + fix FCnt leakage + normal-only Isolation Forest + node-level detection.

That's enough to handle a large portion of tomorrow's questions confidently.